

DBS Chapter 1 – Databases

Questions and Answers

Section A – Basics

1. Define Database and DBMS with examples

Database: Organized collection of related data.

DBMS: Software used to create, manage, and access databases.

Examples: MySQL, Oracle.

2. What is Metadata?

Metadata is data about data. It describes structure, data types, and constraints.

3. Difference between Database and Database System

Database is collection of data.

Database System is Database + DBMS software.

4. What is a Transaction? Give example

A transaction is a single logical operation.

Example: Bank transfer.

5. What is Data Redundancy?

Same data stored multiple times causing inconsistency and storage waste.

Section B – Concepts

6. Explain File System vs DBMS

File system has redundancy and poor security.

DBMS has centralized storage, better security, and consistency.

7. Explain Characteristics of Database Approach

Self-describing nature, data independence, multiple views, and multiuser sharing.

8. What is Data Independence?

Ability to change database structure without changing programs.

9. Explain Multiple Views of Data

Different users see different subsets of database.

10. What is Concurrency Control?

Allows multiple users to access database safely without conflict.

Section C – ACID & Integrity

11. Explain ACID Properties

Atomicity: All or nothing

Consistency: Database remains valid

Isolation: Transactions independent

Durability: Data saved permanently

12. What are Integrity Constraints?

Rules to ensure correct data like unique keys and foreign keys.

13. Explain Referential Integrity

Foreign key must match primary key in another table.

14. Why are Indexes Important?

Indexes speed up searching in database.

Section D – Advantages & Disadvantages

15. Advantages of DBMS

Less redundancy, better security, backup, multiuser access, efficient queries.

16. Situations where DBMS is not Suitable

Small embedded systems, single-user apps, real-time strict systems.

17. How DBMS Provides Security?

User authentication, authorization, and access control.

18. What is Backup and Recovery?

Backup copies database. Recovery restores after failure.

Section E – Long Questions

19. Explain Database System Architecture Basics

Includes database, DBMS software, users, applications, and hardware.

20. Importance of DBMS in Modern Society

Used in banking, hospitals, universities, and social media systems.

21. Compare File System and DBMS

File systems have redundancy and inconsistency.

DBMS provides centralized storage, security, and efficient management.